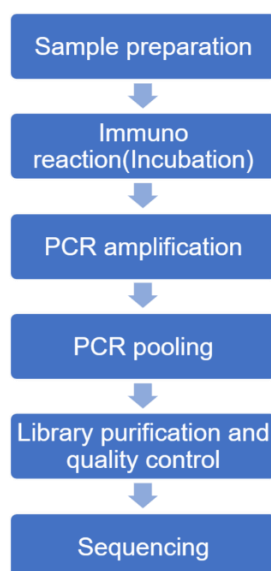


Method

1 Experimental Procedure

Samples and controls are transferred to a sample plate. High-multiplex matched pairs of antibodies, labeled with unique DNA oligonucleotides, bind to their respective proteins in the samples. Proximal oligonucleotides hybridize and undergo extension by Polymerase Chain Reaction (PCR). Final DNA amplicons include specific barcode sequences for each assay and sample specific indexes. The libraries are subsequently pooled and undergo purification using magnetic beads before sequencing on NovaSeq 6000.



2 Bioinformatics Analysis Pipeline

2.1 Data Normalization and Quality Control

NPX, Normalized Protein eXpression, is Olink's arbitrary unit which is in Log2 scale. Following sequencing, the raw data is converted to counts, assigning an integer value to each assay-sample combination based on the detected copy numbers. These raw data counts are converted into NPX values, enabling the identification of protein level changes within different sample sets, and establishing protein signatures. In general, a higher NPX value refer to a higher protein concentration.

2.2 Differential Expression Analysis

The protein expression difference is calculated using standardized NPX data, employing the t-test or ANOVA method under different conditions. The t-test is used for comparing a single variable between two groups, while ANOVA is employed when covariates exist, and more than two groups are being compared. For each method, the analysis is performed separately using the function `olink_ttest` and `olink_anova` from the R package `OlinkAnalyze` (Olink Proteomics AB, 2023). Significant differentially expressed proteins are then filtered using adjusted p-values.

2.2.1 `olink_ttest`

`olink_ttest` performs a t-test for a given grouping variable as defined by variable argument. Paired t-test can be performed by defining `pair_id` argument. Variable must have 2 groups with more than 2 samples per group.

2.2.2 `olink_anova`

`olink_anova` performs an ANOVA F-test for each protein based on variables and covariates defined by variable and covariate arguments or by formula defined by `model_formula` argument [1].

2.3 Enrichment Analysis

Gene Ontology (GO) enrichment analysis of differentially expressed genes were implemented by the clusterProfiler R package, in which gene length bias was corrected. GO terms with corrected P-value less than 0.05 were considered significantly enriched by differential expressed genes. KEGG is a database resource for understanding high-level functions and utilities of the biological system, such as the cell, the organism and the ecosystem, from molecular-level information, especially large-scale molecular datasets generated by genome sequencing and other high-through put experimental technologies (<http://www.genome.jp/kegg/>). We used clusterProfiler R package to test the statistical enrichment of differential expression genes in KEGG [2] pathways. The Reactome database brings together the various reactions and biological pathways of human model species. Reactome pathways with corrected P-value less than 0.05 were considered significantly enriched by differential expressed genes.

3 Reference

- [1] Olink Proteomics AB. (2023). Facilitate Analysis of Proteomic Data from Olink with Olink® Analyze. Retrieved from <https://olink.com/content/uploads/2023/05/olink-analyze-cheatsheet.pdf>
- [2] Kanehisa M, Goto S. KEGG: kyoto encyclopedia of genes and genomes[J]. Nucleic acids research, 2000, 28(1): 27-30.